FROM HAND TO HAND: A Shared Wearable Interface and Functional Benchmark for Dexterous Manipulation

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Abstract: Collecting dexterous manipulation demonstrations across robot hands often requires embodiment-specific sensing hardware or human-motion retargeting. We present WUDI, a Wearable, Unified Interface for Dexterous manipulation that separates a shared sensorized exoskeleton from interchangeable passive hands and linkages. Its 16 sensors remain on the exoskeleton, while a mechanism model reconstructs target-hand joint positions. Wrist images and poses enable robot-free demonstration collection. We also introduce OTHER, an eight-task benchmark inspired by occupational therapy for evaluating functional manipulation. We fabricate WUJI and LEAP configurations, assess usability with 12 participants using WUJI, and validate joint reconstruction on the LEAP thumb and index finger against motion capture. Separate policies trained from 70 demonstrations per task–hand pair achieve 15%–80% success across three tasks, with 20 rollouts per pair. For one expert operator using LEAP, WUDI provides higher collection throughput than vision-based teleoperation. These results support shared sensing across embodiments while exposing limitations in reachable motion, contact-force regulation, and passive-hand mechanics. Project Website: https://wudiwudiwudi2026.github.io/

1 Introduction

Everyday manipulation involves more than picking and placing objects: opening a bottle, pulling a zipper, or inserting a buckle requires coordinated finger motion, sustained contact, and controlled force. Achieving this class of behavior with robot hands remains an open problem, and two practical bottlenecks hold back progress. The first is data: learning these behaviors from real-world demonstration requires large amounts of expert data suited to the target robot hand. The second is evaluation: without a common task suite, it is difficult to say whether a new hand, interface, or policy represents progress over prior work. Differences in joint structure, link geometry, and available motion complicate direct demonstration transfer across robot hands and make comparisons difficult when evaluation protocols also differ.

Mechanically coupled interfaces [1, 2, 3] combine sensing with embodiment-specific structure, so adapting them to a new hand can require redesign beyond passive components. Evaluation also suffers from differing task protocols. Shared tasks and human-operated references can contextualize autonomous performance, although they do not independently isolate hardware and learning limitations.

Our key insight has two parts. First, a generalizable data-collection interface should separate sensing hardware from embodiment-specific geometry, and should provide sufficient degrees of freedom (DoF) to accommodate different hands. Without both, adapting an interface to a new hand can require redesigning the sensorized components and exoskeleton, rather than only the passive parts. Second, the problem of comparing functionally different hands on shared tasks is not new. Occupational therapy (OT) has developed validated instruments for exactly this. The Sollerman Hand Function Test [4] scores twenty activities of daily living: what makes it portable is that it is grounded in functional outcome rather than in a particular anatomy. We argue that the same property makes it

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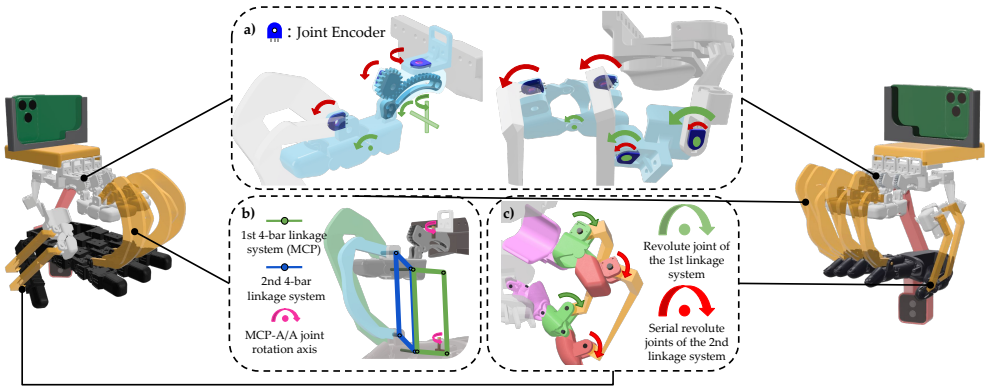

Figure 1: WUDI uses a common sensorized exoskeleton (blue encoder locations) with hand-specific passive replicas and linkages for LEAP and WUJI. Two serial four-bar couplers transmit non-thumb MCP and PIP motion.

a natural basis for measuring robot dexterity. Supporting motion is not the same as being dexterous, and the available motion must be shown to enable functional manipulation.

We present WUDI, a wearable interface that implements this separation through a sensorized exoskeleton, interchangeable passive hands, and embodiment-specific linkages. Adapting the interface to a new hand requires selecting coupler dimensions and checking mechanical compatibility, and retains the sensing hardware unchanged. Fixed or coupled joints can reduce independent degrees of freedom and reachable workspace. To assess functional dexterity across embodiments, we introduce the Occupational Therapy-inspired Hand Evaluation and tRaining (**OTHER**) benchmark. OTHER combines activities adapted from the Sollerman Hand Function Test with busy-board tasks that consolidate the fastening, latching, and rotation activities of OT practice into a single reproducible fixture. It evaluates both human-operated demonstration collection and autonomous execution, so that the performance reachable by a human teleoperator on a given hand serves as a reference for what a policy on that hand could achieve.

2 Wearable Interface Design

Shared exoskeleton. The exoskeleton carries every encoder, the acquisition board, and all wiring, and is worn independently of the target robot hand. Each of four non-thumb fingers provides metacarpophalangeal (MCP) abduction/adduction and flexion/extension plus proximal interphalangeal (PIP) flexion. Three further motions accommodate the thumb base, MCP, and interphalangeal joints: 15 operator-side degrees of freedom in total, sensed by 16 rotary encoders. Some encoders measure coupler rotation rather than a hand joint directly. The common electronics therefore need not fit inside a new passive hand; the trade-off is that the passive angles have to be inferred from the mechanism. The shared electronics and one printed hand cost about $120 in listed parts, excluding the wrist camera and phone.

Passive hand and coupling. The passive replica retains the target's supported finger layout, joint axes, and contact shape (Fig. 1). The source hand's meshes and kinematic description are normalized and exported as an articulated STEP assembly. Joint cores, linkage mounts, collision relief, and stops then receive reviewed CAD edits. Two serial four-bar linkages per non-thumb finger carry MCP and PIP flexion: the first coupler connects PIP axes, and its moving segment serves as the ground for the second linkage. Standoffs set the relative exoskeleton and passive-hand bases. Because the passive fingers can be inclined relative to the exoskeleton, flexion may induce passive-hand abduction; we model the complete spatial mechanism rather than equating the two hands' encoder angles. This embodiment-specific geometry resides in printed components and their model, while the electronics remain fixed.

Thumb and reachable motion. Thumb axes vary more strongly between hands, so their couplers and attachment points are selected per embodiment. WUJI keeps four passive-thumb joints movable, coupled to the exoskeleton's three thumb coordinates. In the evaluated LEAP configuration, the thumb-root coordinates are fixed to prioritize MCP and IP flexion. Available distal joints are held at 30° where required by the linkage. Each fixed or coupled coordinate uses the same constraint during demonstration and robot execution. These choices preserve direct correspondence over a useful task range, while reducing the reachable workspace; they are visible in the coverage analysis below.

Calibration and reconstruction. Per-channel affine calibrations map sensor counts to measured exoskeleton or coupler angles. The exoskeleton, passive hand, and bars are assembled into one kinematic model. Direct measurements constrain the exoskeleton coordinates, while bounded nonlinear least squares fit passive joint angles and bar poses to pin-position, axial, and angular closure residuals. Warm starts and a continuity penalty favor a consistent trajectory; large residuals or implausible jumps are rejected. The same reference configuration and joint-axis convention convert the inferred passive displacements into target-robot commands. Appendix A gives the kinematic construction, and Appendix C reports per-joint errors.

Portable observations. A wrist-mounted RealSense D405 records RGB-D, and a keyed iPhone mount supplies session-relative six-degree-of-freedom ARKit poses. Both are carried by the operator, enabling collection away from the robot. The keyed mount maintains the phone-to-palm transform; recordings synchronize camera, phone, and encoder streams while retaining temporal offsets. The camera is placed at corresponding wrist viewpoints during collection and execution (Fig. 2). In our policy experiments, this arrangement supports use of the recorded RGB frames as observations without segmenting or replacing the passive hand. It does not guarantee appearance matching for arbitrary external views or robot geometries.

Figure 2: Matched wrist views with passive and robot hands.

3 Evaluation

OTHER comprises Zipper, Spin, Lid, Write, Coin, Buckle, Squeeze, and Unbuckle. We score the object-level outcome and evaluate autonomous policies on Zipper, Lid, and Coin with both hands.

User study. In an approved study, 12 participants used the WUJI configuration on eight tasks after five minutes of practice. They made up to five attempts per task and completed five-point ratings. Of 480 planned attempts, 397 were recorded; participants could skip remaining attempts after a timeout, and the 83 unrecorded slots were excluded from success denominators. Further protocol details are in Appendix B.

Success was 60/60 for each of Zipper, Spin, and Lid; 56/60 for Write, 52/60 for Coin, and 44/53 for Buckle. No participant succeeded on Squeeze (0/23) or Unbuckle (0/21). Squeeze was limited by pinch force and lack of tactile sensing, while Unbuckle exposed the passive hand's dimensions and structural strength. Donning and doffing took 16.81 ± 6.27 and 4.05 ± 3.98 seconds (mean $\pm$ SD); perceived success averaged 3.83/5 and willingness for extended collection 3.08/5. These results describe the tested WUJI configuration under an optional-continuation protocol; full plots are in Appendix B.

Autonomous policies. With each hand on an xArm 6, we collected 70 demonstrations per task–hand pair and trained separate Diffusion Policies [5]. A ResNet-18 and temporal U-Net map wrist RGB and relative end-effector pose to relative arm actions

Table 1: Policy successes in 20 trials.

Hand	Zipper	Lid	Coin
WUJI	16/20	15/20	9/20
LEAP	11/20	3/20	13/20

and absolute hand targets. Table 1 gives success in 20 two-minute rollouts per pair. On LEAP–Lid, excessive closure ejected the lid in 16 of 17 failures. An expert completed this task with the same hand; position-only data leave grip-force regulation unresolved.

Reconstruction accuracy. Motion capture of the LEAP passive thumb and index finger gives mean absolute joint errors of $1.34°$–$4.20°$ and root-mean-square errors at most $5.50°$ over the sampled motion range. Index MCP flexion has the largest mean error; the per-joint results appear in Appendix B. This check supports the inferred joint targets on LEAP, but does not measure reconstruction accuracy for WUJI or all possible poses.

Collection throughput. The vision baseline uses HaMeR [6] on multiview RGB to estimate human-hand pose, retargeting it to LEAP for real-time teleoperation. With one experienced operator and five five-minute windows per task and condition, WUDI averaged 15.0 versus 2.0 successful Zipper completions, 9.4 versus 2.6 Coin completions, and 11.0 versus 1.4 Lid completions per window (Fig. 3). These are 7.50, 3.62, and 7.86 times the baseline counts. Two-sided exact permutation tests on window counts with Holm correction yielded adjusted $p = 0.0238$ for each task. Bare-handed trials provide a human reference but no robot-executable data. Throughput across additional operators remains untested.

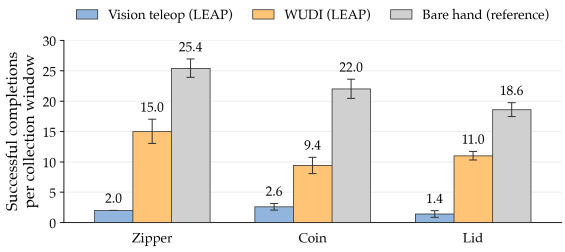

Figure 3: Expert LEAP completions per five-minute window. Bars show means ± SD over five windows; bare-handed trials are a reference.

	LEAP		WUJI	
	Thumb	Non-thumb	Thumb	Non-thumb
WUDI / Vision	0.27	0.84	0.83	0.58

Figure 4: Native, WUDI, and vision-teleoperation fingertip workspaces for both hands. Values are reconstructed-volume ratios, WUDI/vision.

Workspace coverage. WUDI's fixed and coupled coordinates reduce fingertip motion relative to vision teleoperation (Fig. 4). The reconstructed-volume ratios are 0.27 for the LEAP thumb, 0.84 for LEAP non-thumb fingers, 0.83 for the WUJI thumb, and 0.58 for WUJI non-thumb fingers. LEAP's thumb-root constraint produces the strongest loss. These volumes describe motion coverage rather than task success; a smaller region can still contain the grasps needed for the evaluated tasks.

Across-hand outcomes. WUJI policies succeeded more often on Zipper (16/20 versus 11/20) and Lid (15/20 versus 3/20), while LEAP succeeded more often on Coin (13/20 versus 9/20). The results show that the same collection interface supports different task strengths across hands. They combine embodiment geometry, collected trajectories, and policy learning, so the observed gaps do not isolate hardware capability.

4 Discussion

The shared sensing assembly supported collection and policy training on both hands, with different consequences for each. The LEAP–Lid failures show that joint-position demonstrations do not convey the grip force needed after contact. The WUJI study exposed limits in pinch force and passive-hand strength, while the workspace comparison makes the cost of mechanical coupling visible. Broader claims require testing more hands, operators, and contact-rich tasks; the five additional designs in Appendix E have only CAD and simulation checks.

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A Design and Implementation Details

A.1 System architecture and fabrication

WUDI has three mechanical parts: a sensorized wearable exoskeleton, a passive replica of the target robot hand, and embodiment-specific couplers between them (Fig. 5). The exoskeleton and its electronics are reused; the printed hand, linkage geometry, and mechanism model change with the robot. The passive replica preserves the target's supported joint axes, finger arrangement, and contact geometry, although the linkage may fix or couple some coordinates. These choices establish mechanical correspondence only inside the supported motion range.

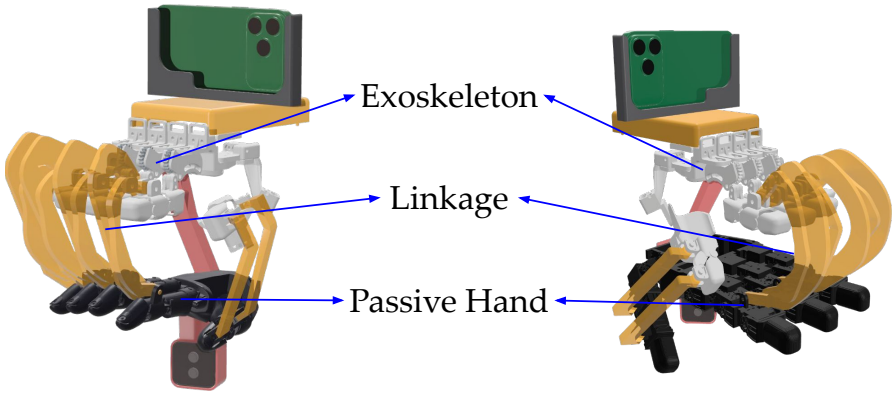

Figure 5: Exoskeleton, couplers, and passive hand in the WUJI (left) and LEAP (right) assemblies. The sensors and wiring stay on the upper assembly when the passive hand is swapped.

For each of four non-thumb fingers, the exoskeleton supports MCP abduction/adduction (AA), MCP flexion/extension (FE), and PIP flexion. Its thumb accommodates CMC, MCP, and IP rotations, giving 15 operator-side coordinates in total. Sixteen resistive rotary sensors measure exoskeleton joints or a coupler's rotation relative to its supporting segment. Lateral clearance on the index, middle, and ring fingers prevents direct coaxial MCP-FE shafts; arc-slot transmissions carry those rotations to accessible shafts, following ART-Glove [7]. The little finger allows a coaxial sensor. The acquisition PCB connects ALPS RDC506018A sensors to two ADS1256 ADCs and an STM32 microcontroller.

To construct a new passive hand, the source robot's link meshes and kinematic description are normalized to a common assembly reference. Joint axes and ordering are retained, and a named, faceted STEP assembly is exported through FreeCAD and checked for solid integrity. Joint cores, pin holes, linkage mounts, interference relief, and mechanical stops are then completed with reviewed CAD edits. These steps require embodiment-specific mechanical judgment; generating the mesh assembly does not establish linkage feasibility by itself.

A.2 Finger placement and linkage closure

Two serial four-bar linkages transmit non-thumb MCP and PIP flexion, following the coupled-hand principle used by DEXOP [1]. Standoffs set the relative placement of the exoskeleton and passive-hand bases. For each embodiment, a fore–aft offset x and vertical separation h determine the reference geometry and pin-to-pin coupler lengths. The first coupler connects the two PIP axes and supplies the second linkage's moving ground; the second connects middle-link mounts with matching offsets in their respective PIP frames. Revolute pins close the mechanism (Fig. 1).

The passive fingers retain their native orientations. In an inclined layout such as WUJI, coupling to parallel exoskeleton fingers can induce passive-hand AA motion during FE. Joint reconstruction therefore solves the spatial mechanism and does not equate the exoskeleton and passive-hand

abduction angles. Where the target has a distal interphalangeal (DIP) coordinate that this linkage cannot drive independently, it is fixed at 30° in both the replica and the robot during the evaluated demonstrations and rollouts.

Thumb layouts vary more than the non-thumb layouts. Two couplers connect mounts above the exoskeleton's thumb segments to the passive thumb; their locations, axes, and lengths are chosen manually for each hand while checking grasping range and collision clearance. WUJI retains four movable passive-thumb joints coupled to the exoskeleton's three thumb coordinates, preserving metacarpal reorientation without one-to-one correspondence. The evaluated LEAP adaptation prioritizes MCP and IP flexion and fixes both thumb-root coordinates. That choice reduces the LEAP thumb's workspace, as quantified in Fig. 4.

A.3 Calibration, reconstruction, and recording

The exoskeleton, passive hand, and bars form one MuJoCo kinematic model. Registering the passive assembly to the source robot establishes its reference configuration, axes, and joint ordering. For a reference-frame joint axis through point $\mathbf{c}_j$ with unit direction $\mathbf{a}_j$ and displacement Δq_j, its transform and a descendant link pose are

$$\begin{aligned} G_j(\Delta q_j) &= \begin{bmatrix} R_j & (I - R_j)\mathbf{c}_j \\ \mathbf{0}^\top & 1 \end{bmatrix}, \\ T_{WL}(\Delta \mathbf{q}) &= \left(\prod_{j \in \mathcal{A}(L)} G_j(\Delta q_j) \right) T^0_{WL}, \end{aligned} \tag{1}$$

where $R_j = \mathrm{Rot}(\mathbf{a}_j, \Delta q_j)$, T^0_{WL} is the assembly-reference pose, and $\mathcal{A}(L)$ is the ordered root-to-link joint path. The same forward model propagates attachment points and bar poses so that measured and inferred states can be checked against linkage closure.

Per-channel affine calibration converts ADC counts to exoskeleton angles or carrier-relative coupler angles, accounting for sensor direction, offset, and transmission gain. Geared MCP sensors use mechanical stops 50° apart. Sensor calibration remains with the reusable exoskeleton; a reference-pose capture registers each mounted passive assembly. Direct measurements fix the observed exoskeleton coordinates. Bounded nonlinear least squares then infers passive joint angles, remaining coordinates, and rigid-bar poses from mount-angle errors and radial, axial, and angular pin residuals, allowing finite assembly clearance. Fixed joints retain their specified angles. Warm starts and continuity regularization stabilize the trajectory; large residuals or increments trigger rejection. Residual and Jacobian diagnostics record constraint consistency and local ambiguity.

Encoder, phone, and RGB streams are recorded independently against a shared clock. A Hampel filter followed by a One Euro filter [8] smooths the calibrated encoder signals before reconstruction. Inferred displacements are converted to absolute robot-hand targets using reference angles and axis conventions. Camera and phone samples are matched to trajectory timestamps by nearest neighbor, retaining offsets and validity flags. Raw streams remain available for reprocessing.

The keyed wrist mount maintains the phone-to-palm extrinsic transform. An iPhone running Record3D supplies session-relative six-degree-of-freedom ARKit poses over USB; a RealSense D405 supplies wrist RGB-D frames. Figure 6 shows the acquisition interface's visual previews, pose trace, and per-joint readings. The listed parts for the common sensing assembly and one printed hand, including sensors, electronics, PCB, and one spool of PLA-CF, cost about $120. The D405 and phone are excluded. Additional hand configurations add printed parts and fasteners.

B OTHER Tasks and Study Protocol

B.1 Functional task definitions

OTHER (Occupational Therapy-inspired Hand Evaluation and tRaining) uses eight tasks spanning precision grasp, tool use, pulling, rotation, compression, and fastening. Coin, Lid, and Write adapt

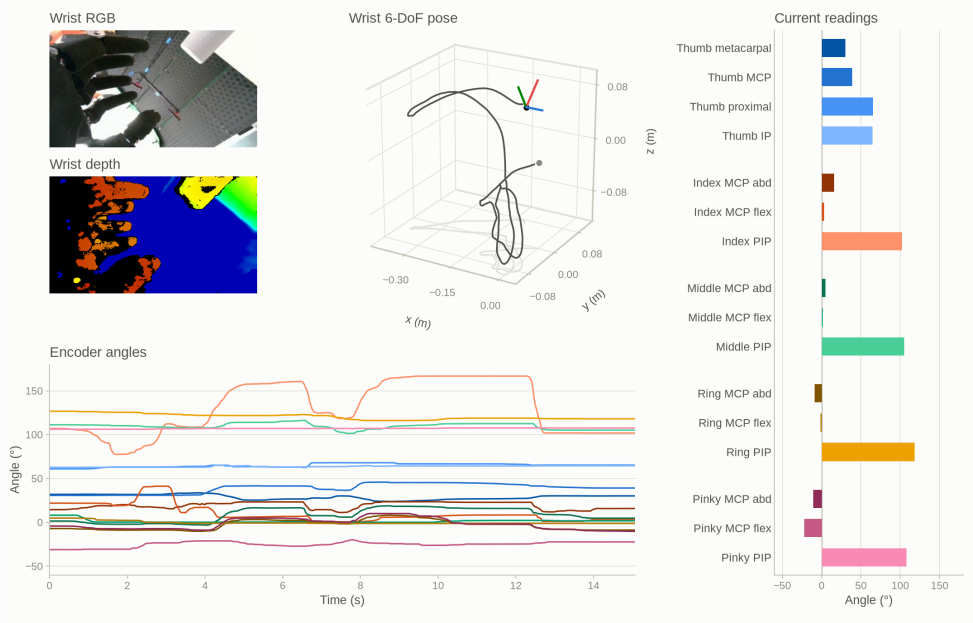

Figure 6: Acquisition display with wrist RGB and depth, session-relative wrist pose, encoder traces, and current joint readings.

activities in the Sollerman Hand Function Test [4]; Zipper, Spin, Squeeze, Buckle, and Unbuckle use common busy-board mechanisms. We use object-level outcomes, allowing distinct hands to adopt different motion strategies. Table 2 gives the completion condition and principal demand of each task. These are robotic task outcomes, not the clinical test's scoring scale.

Table 2: OTHER tasks, completion conditions, and principal dexterity demands.

Task	Completion condition	Dexterity demand
Coin	Pick up a coin and deposit it in a jar.	Precision grasp and release
Lid	Fully unscrew a bottle lid.	Grip–torque coordination
Write	Write an "S" with a marker.	Tool grasp and contact control
Zipper	Pull a slider to the end of its track.	Directionally constrained pulling
Spin	Rotate a wheel by 180°.	Rotational control
Squeeze	Compress an object until it sounds.	Forceful compression
Buckle	Insert and latch a buckle.	Alignment and insertion
Unbuckle	Release the buckle latch.	Coordinated latch actuation

We evaluate human-operated performance on all eight tasks. Zipper, Lid, and Coin were selected for autonomous policies as representative tasks from the observed easier, intermediate, and harder groups, respectively. Squeeze and Unbuckle were not used for policy learning because operators did not obtain successful demonstrations with this configuration.

B.2 Human-operated study

An IRB-approved study evaluated the WUJI configuration with 12 participants (six women and six men). Hand lengths were 15.5–19.9 cm and widths were 8.0–11.0 cm. Each participant first attempted every task once bare-handed. We then measured donning time, allowed five minutes of free practice, and asked for five consecutive trials per task in randomized task order (40 planned trials per participant). A trial ended at success, timeout, participant request, or an unrecoverable state. After a timeout, participants could retry or move to the next task. They later provided five-point questionnaire ratings and qualitative feedback.

Figure 7: Representative OTHER tasks performed bare-handed (upper rows) and through the WUJI configuration of WUDI (lower rows). The collage shows task objects and hand approaches; the scored outcomes are defined in Table 2.

The optional-continuation protocol yielded 397 recorded attempts out of 480 planned slots. The remaining 83 were unrecorded, rather than observed failures, and are excluded from the success-rate denominators in Table 3. Uncertainty estimates use participant-level bootstrap resampling, preserving all of a sampled participant's trials. Completion-time summaries exclude unsuccessful trials and first average successful trials within each participant and task.

Table 3: Recorded WUJI user-study attempts; missing planned slots are excluded. The difficulty grouping describes this interface under the optional-continuation protocol.

Task	Successes	Attempts	Success rate	Observed group
Zipper	60	60	100%	Easy
Spin	60	60	100%	Easy
Lid	60	60	100%	Medium
Write	56	60	93.3%	Medium
Buckle	44	53	83.0%	Hard
Coin	52	60	86.7%	Hard
Squeeze	0	23	0%	Challenging
Unbuckle	0	21	0%	Challenging

Buckle had the longest mean successful completion time, 19.21 s (participant-bootstrap 95% CI: 13.13–26.00 s). Donning took 16.81 ± 6.27 s and doffing 4.05 ± 3.98 s (mean $\pm$ SD). Perceived

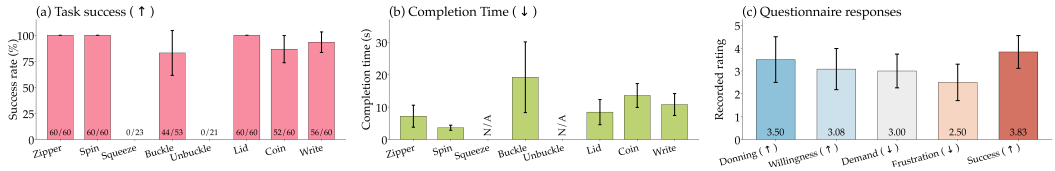

Figure 8: User study ($n = 12$): (a) success among recorded attempts; (b) successful completion times, averaged within participant and then shown as mean $\pm$ sample SD across participants (N/A when nobody succeeded); (c) questionnaire responses as mean $\pm$ sample SD. Demand and frustration remain in their original direction, not reverse coded.

success averaged 3.83/5 and willingness for an extended collection session 3.08/5. Participants' success on six tasks supports practical demonstrations within this WUJI workspace, while the willingness rating and two consistently failed tasks mark user-experience and mechanical limitations. Squeeze was limited by pinch force and the lack of tactile sensing; Unbuckle exposed passive-hand dimensions and structural strength. The low recorded-attempt counts on these two tasks require the optional-continuation caveat when interpreting the zero success rates.

C Policy and Measurement Protocols

C.1 Policy collection, actions, and rollouts

The two physical configurations use WUJI Hand v1 and LEAP Hand, each mounted on a UFactory xArm 6 for policy evaluation. For each task–hand pair, 70 demonstrations were recorded with WUDI and a separate Diffusion Policy [5] was trained. The observation contains wrist RGB and the relative arm end-effector pose. The action contains relative end-effector motion and absolute target-hand joint positions reconstructed from exoskeleton readings. The visual encoder is ResNet-18 and the action model is a temporal U-Net. Recorded observations are used without hand segmentation, inpainting, or embodiment-specific image replacement. Each of the six trained policies is evaluated for 20 rollouts with a two-minute timeout; task success follows Table 2.

The WUJI policies succeeded in 16/20 Zipper, 15/20 Lid, and 9/20 Coin rollouts; LEAP succeeded in 11/20, 3/20, and 13/20, respectively (Table 1 in the main paper). LEAP's bulkier fingers can facilitate the Coin pinch, whereas WUJI's smaller geometry may hinder full closure for operators with larger hands. These observations do not isolate a hardware-only effect because the demonstrations and trained policies also differ. Of LEAP's 17 Lid failures, 16 involved excessive closure after contact and ejection of the lid. An expert could complete Lid with LEAP in the collection comparison; the low policy rate is therefore not evidence that the task is unreachable with that hand. WUDI measures positions without grip forces, leaving contact-force modulation to inference from the policy observations.

Figure 9 shows representative autonomous sequences for WUJI and LEAP hand on Zipper, Lid, and Coin tasks. Quantitative results for all six task–hand pairs are reported in Table 1.

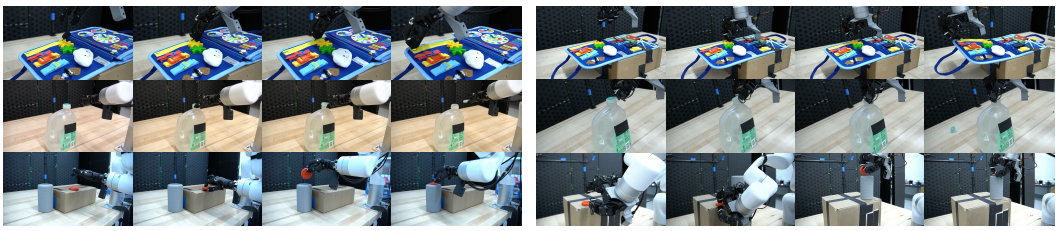

Figure 9: Example autonomous sequences. Left three rows: WUJI Zipper, Lid, and Coin. Right three rows: LEAP Zipper, Lid, and Coin.

C.2 Joint reconstruction accuracy

Passive-hand joint reconstruction was compared with motion-capture markers on the LEAP thumb and one representative non-thumb finger (index); the three LEAP non-thumb fingers use the same linkage structure. Ground-truth joint angles from markers were compared to WUDI's inferred angles over the sampled motion range. Table 4 gives mean absolute error (MAE), root-mean-square error (RMSE), and maximum absolute error. The index MCP-FE had the largest mean error, $4.20°$; measured maxima ranged from $9.27°$ to $14.06°$.

Table 4: LEAP passive-hand joint reconstruction errors in degrees.

Finger	Joint	MAE	RMSE	Max. abs.
Thumb	MCP	3.17	3.97	10.09
Thumb	IP	3.33	4.55	12.84
Index	MCP-AA	2.10	2.89	10.23
Index	MCP-FE	4.20	5.50	14.06
Index	PIP	1.34	2.33	9.27

C.3 Fingertip workspace and collection throughput

For both robot hands, fingertip regions obtained under native articulation, WUDI's mechanical constraints, and vision-teleoperated robot commands were reconstructed from sampled positions. The calculation uses Delaunay tetrahedra with circumsphere radii no larger than 25 mm and rasterizes the resulting regions at 2 mm resolution. The vision commands are produced by retargeting human motion within the robot's joint limits without the physical constraints of WUDI. Ratios of WUDI to vision reconstructed volume are 0.27 (LEAP thumb), 0.84 (LEAP non-thumb), 0.83 (WUJI thumb), and 0.58 (WUJI non-thumb), as plotted in Fig. 4. The LEAP configuration locks both thumb-root coordinates and fixes available non-thumb DIP joints at $30°$; vision teleoperation imposes neither restriction. Reconstructed-volume coverage describes sampled position regions, not every reachable contact orientation.

The vision-teleoperation throughput baseline applies HaMeR [6] to multiview RGB, estimates the operator's 3D hand pose, and retargets it to LEAP joints online. LEAP was used for both interfaces because its compliance tolerates some pose-estimation error. One experienced operator performed five five-minute windows per task and condition on Zipper, Coin, and Lid, with bare-handed operation as a human-performance reference. The per-window completion means appear in Fig. 3; error bars are sample SD across the five windows. Two-sided exact permutation tests compare window-level counts, with Holm correction over the three tasks; each adjusted value is $p = 0.0238$. The observed throughput advantage applies to this operator and setup.

D Comparison with Prior Interfaces and Benchmarks

Mechanical correspondence. UMI [9] provides portable robot-free gripper demonstrations; DexUMI [3] extends wearable collection to a dexterous hand with visual correction. DEXOP [1] and DexEXO [2] use passive hand replicas to align operator motion and contact geometry with a particular robot; RealDexUMI [10] uses an isomorphic glove. These devices establish useful correspondence but are presented for particular hand platforms. The distinction in WUDI is the location of its sensors: the exoskeleton retains all instrumented components when its passive hand and linkages are changed. Table 5 summarizes the design properties used in the original comparison.

Human-motion mapping. AnyTeleop and TeleVision [11, 12] recover hand motion with cameras, while DOGlove and ART-Glove [13, 7] use instrumented wearables; DexCap [14] collects motion without the robot. Mapping human motion to a robot then has to accommodate differing joint structures, limits, and contacts. Related approaches include trajectory retargeting [15],

object-guided learning [16], learned shared intent [17], residual learning [18], and physics-informed retargeting [19]. WUDI uses its coupled mechanism to provide robot-shaped contact and infer target joint motion in the supported configuration range; it still needs per-hand coupling design and mechanism reconstruction.

Table 5: Comparison of data collection interfaces.

Method	No Vis. Adapt.	Diff. Hand	Port.	No Retarg.	Full ROM
Mechanically Coupled / Isomorphic					
DexUMI [3]	×	×	✓	✓	×
RealDexUMI [10]	✓	×	✓	✓	×
DEXOP [1]	✓	×	✓	✓	×
DexEXO [2]	✓	×	✓	✓	×
Human-Motion Capture / Glove-Based					
DexCap [14]	×	○	✓	×	✓
AnyTeleop [11]	✓	✓	×	×	✓
TeleVision [12]	✓	✓	×	×	✓
DOGlove [13]	✓	○	×	×	×
OSMO [20]	✓	○	✓	×	✓
ACE [21]	✓	✓	×	×	✓
WUDI	✓	✓	✓	○	×

No Vis. Adapt.: no embodiment-gap correction of visual observations, such as inpainting or hand-geometry replacement. *Diff. Hand*: reuse of the sensing interface across different robot hands; replacing embodiment-specific passive parts is allowed. *Port.*: portable collection without a robot arm. *No Retarg.*: no human-to-robot hand-motion retargeting, excluding calibration of joint encoder. *Full ROM*: Hand motion is not mechanically constrained. ✓ = yes, ○ = limited or not established, × = no.

Functional benchmarks. NIST Assembly Task Boards [22] and Robothon/euROBIN [23] standardize physical assembly tasks, generally centered on arm-level manipulation and parallel-jaw grippers. Simulation suites provide controlled task and embodiment variation [24, 25, 26, 27, 28] but cannot fully expose backlash, actuation uncertainty, and contact compliance. POMDAR [29] evaluates physical tasks across ORCA hand embodiments with teleoperation. OTHER draws on the functional-outcome perspective of the Sollerman assessment [4] and permits comparison between human-operated demonstrations and learned robot policies on the same object-level tasks.

E Additional Designs, Qualitative Examples, and Limits

Five more passive-hand and coupler designs were assembled in CAD and checked for linkage closure in MuJoCo: ROHand, Inspire Hand, Sharpa Wave, XHand 1, and Schunk Hand (Fig. 10). These designs require mount adjustments and embodiment-specific coupler dimensions. They were not fabricated or used for demonstration collection because the corresponding physical hands were unavailable. The physically evaluated adaptations in this paper are WUJI and LEAP.

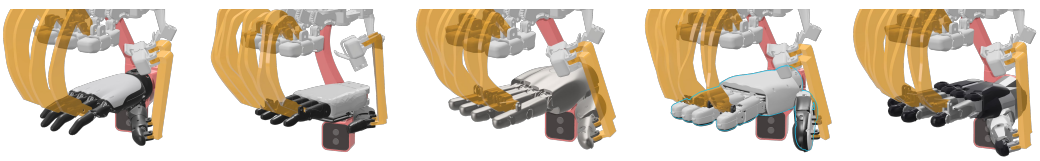

Figure 10: Five additional passive-hand designs that passed CAD and simulation linkage checks. No physical results are claimed for these configurations.

Figure 11 illustrates representative user-study interactions with busy-board and tabletop objects using the WUJI configuration of WUDI.

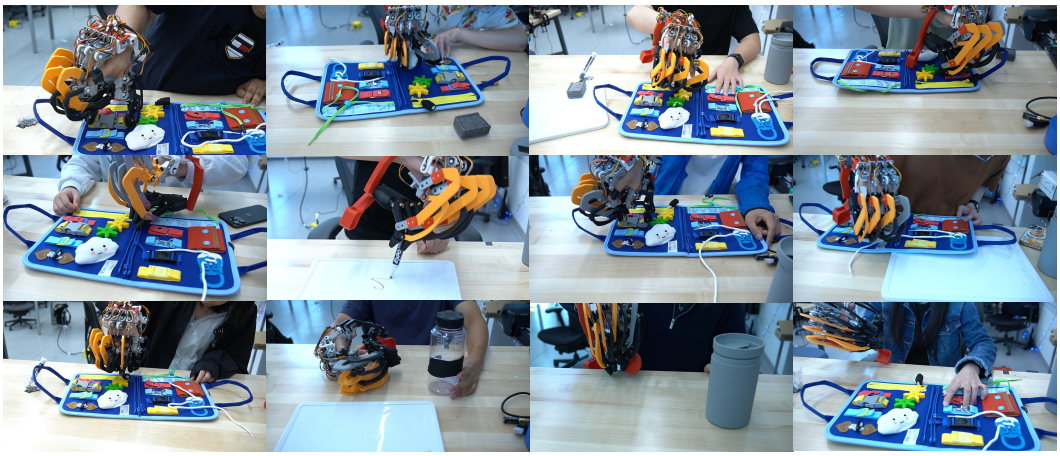

Figure 11: User-study examples with the WUJI passive hand across busy-board mechanisms and tabletop objects.

Several constraints remain. Fixing or coupling joints can substantially reduce articulation, particularly on the LEAP thumb, and thumb mounts still require manual CAD work. The exoskeleton and linkage structure can interfere in confined spaces. Wrist-view correspondence does not automatically remove the appearance gap in third-person views, which may require visual adaptation. The rigid assembly's fit varies with hand size; concentrated weight and elevated arm posture can reduce comfort during extended collection. Position-only measurements lack contact force, and the current passive-hand pinch force and structural strength prevented completion of Squeeze and Unbuckle in the WUJI study. Work on fit, weight distribution, durability, thumb motion, tactile or force sensing, and measured integration effort is needed to assess broader reuse.